

Intelligent Agents

Assignment: Designing and Evaluating an Intelligent Agent

Chosen Agent: AI Real Estate Recommendation Agent

Property buyers today are often overwhelmed by the sheer number of listings spread across different platforms, making it hard to find a home that truly fits their needs. This assignment presents the design and evaluation of an intelligent agent — the **AI Real Estate Recommendation Agent** — that assists buyers in locating suitable properties. It covers problem identification, problem formulation, PEAS-based evaluation, and classification of the agent's environmental properties, following the standard framework used for analyzing intelligent agents.

1. Identify the Problem

House hunting is usually a slow and frustrating process. A buyer has to manually browse dozens of property listing websites, visit multiple agents in person, and compare prices, locations, and features by hand. This manual approach leads to a few recurring problems:

- **Information overload** — too many listings are available, and comparing them manually takes a lot of time and effort.
- **Mismatched results** — buyers often end up viewing properties that don't actually fit their budget, location, or size requirements.
- **Slow decision-making** — without a system to filter and rank options, finding the right property within a reasonable budget and location can take weeks.

Problem Statement

There is a need for an intelligent agent that can automatically collect a buyer's requirements — such as budget, location, and property type — and match them against available property listings, so that the most suitable options are recommended quickly and accurately, without the buyer having to search manually.

2. Formulate the Problem (Text-Based)

The problem is formulated below in terms of the agent's initial state, available actions, goal state, percepts, and the sequence of actions it follows.

Initial State

The agent has no knowledge of the buyer's needs until the buyer provides input — such as budget, preferred location, property type, and number of bedrooms required.

Actions Available to the Agent

- Search the property listings database
- Filter listings by budget, location, and property type
- Rank the filtered results by suitability

- Present the top recommendations to the buyer
- Request clarification if requirements are incomplete

Goal State

The buyer is matched with one or more properties that closely fit their stated budget, location, and requirements, minimizing the time spent searching manually.

Percepts (Inputs the Agent Receives)

- Budget entered by the buyer
- Preferred location
- Property type (house, apartment, plot, etc.)
- Additional filters (number of bedrooms, amenities, etc.)

Sequence of Actions (Path)

Collect buyer requirements → Search listings database → Filter and compare properties → Rank by suitability → Recommend best matches → Refine based on buyer feedback → Repeat

Sample Scenario (Illustrative Data)

Field	Value
Customer Name	Sara
Budget	80 Lakh
Preferred Location	Lahore
Property Type	Apartment
Requirement	2 Bedrooms

Formulated Problem Statement

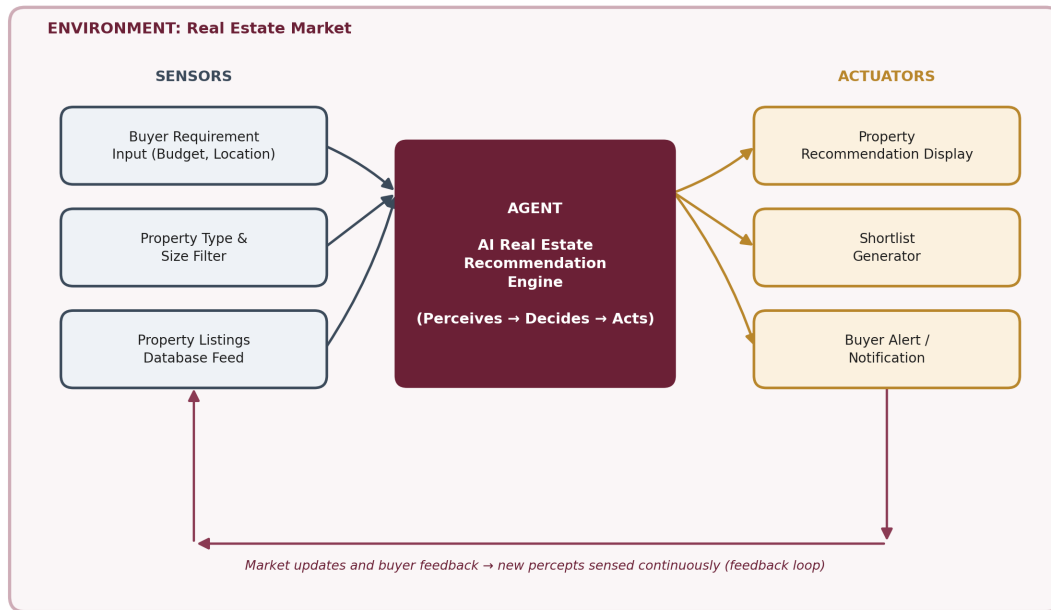
Given the buyer's budget, preferred location, and property type as input, the agent must search and rank the available listings, such that the recommended properties fall within the buyer's stated budget and location, while minimizing the time and effort spent on manual searching.

2.1 Graphical Representation of the Environment

The diagram below illustrates the agent's perception–action cycle within the real estate market environment: sensors gather the buyer's requirements and current listing data, the agent analyzes this information and decides on the best matches, and actuators present the recommendations back to the buyer — whose feedback and the market's own updates then produce new percepts for the next cycle.

Graphical Environment Representation

AI Real Estate Recommendation Agent — Perception-Action Cycle



3. Evaluate in the Context of PEAS

PEAS defines the task environment of an agent through four components: **P**erformance measure, **E**nvironment, **A**ctuators, and **S**ensors.

Component	Details
P — Performance Measure	Accuracy of property matches, speed of recommendation, overall customer satisfaction, and efficiency in narrowing down suitable options.
E — Environment	Buyers, available property listings, and the wider real estate market (prices, locations, availability).
A — Actuators	Property recommendation display, shortlist generator, and buyer alert/notification system.
S — Sensors	Buyer's input (budget, location, property type, filters) and the live property listings database feed.

4. Identify the Environmental Properties

The agent's environment is classified below using the standard set of environment properties.

Property	Type	Justification
Observability	Partially Observable	The agent only knows what the buyer explicitly enters (budget, location, type) — it cannot see unstated preferences or the buyer's true decision-making factors, so its knowledge of the buyer's needs is incomplete.
Determinism	Stochastic	Even when the agent recommends a property, the outcome is not guaranteed — the listing may be sold, the price may change, or the buyer may reject the option for unstated reasons.
Episodes	Sequential	Earlier searches and buyer feedback influence how later recommendations are refined, so each decision is connected to the ones before it rather than being isolated.
Change	Dynamic	The property market keeps changing independently of the agent — new listings appear, prices update, and properties get sold while the agent is still processing a request.
Values	Continuous	Attributes such as price and area are continuous values (e.g., 80.5 Lakh, 1450 sq. ft.), while only some filters like property type are discrete.
Agents	Multi-Agent	Multiple buyers and other agents (or competing platforms) act within the same market simultaneously, competing for the same limited set of properties.

End of Assignment — AI Real Estate Recommendation Agent